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A Leap Towards World Class Education

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Q.1. Define an algorithm. List any four characteristics of an algorithm.

→ An algorithm is a finite sequence of well-defined, step-by-step instruction used to solve a specific problem or perform a computation.

Characteristics :

(1) Input : An algorithm takes zero or more inputs
(2) Output: It produces at least one output.
(3) Definiteness: Each step is clear, precise, unambiguous
(4) Finiteness : The algorithm terminates after a finite number of steps.

Q.2. Define data structure and abstract data type (ADT). Give one example of each.

→ A data structure is a way of organizing, storing and managing data in memory so that

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it can be accessed and modified efficiently.
An abstract Data Type [ADT] is a logical description of a data type that specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples :

- Data structure ⟶ Array
- ADT ⟶ stack [operations : push , pop , peek]

3. Explain the process of converting a problem into a program . Illustrate the role of algorithms in this process.

→ ① Problem Definition : Understand requirements and constraints
② Analysis : Identify inputs , outputs , and processing
③ Algorithm design : Create step-by-step logic to solve the problem.
④ Flowchart / Pseudocode : Represent the algorithm visually or textually.
⑤ Coding : Convert algorithm into a program using a programming language.
⑥ Testing & Debugging : Check correctness and remove errors.
⑦ Documentation & maintenance : Improve and maintain the program.

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Role of Algorithm :—
The algorithm acts as a blueprint of the solution. It ensures correctness, efficiency and make coding easier and error-free.

Q.4. Explain linear and non-linear data structures with suitable examples.

→ Linear Data structure:
Data elements are arranged sequentially
- Array
- Stack
- Queue
- Linked List

Non-linear Data structure:
Data elements are arranged in a hierarchical or interconnected manner.

Example : • Tree
• Graph.

Q.5. Differentiate between algorithm and program. Analyze their roles in SDLC.

→

	Algorithm	Program.
1	step-by step logic	Written code
2	Language independent	language dependent
3	Design level	Implementation level
4	No-syntax rules	Follows syntax rule.

Role in SDLC :

- Algorithm : Used in design phase to plan the solution.
- Program : Used in implementation phase to execute the solution on a computer.

2.6. Examine the relation-ship among data, data structure, and algorithm. Justify how the choice of data structure affects algorithm efficiency.

→ Relationship :

- Data is raw information.
- Data structure organizes the data.
- Algorithm processes the data using data structures.

Justification :

- The efficiency of an algorithm depends on the data structure used.
- searching is an array → slower (O(n))
- Searching in a binary search tree → faster (O(log n))

Conclusion :

Choosing the right data structure improves time and space efficiency of algorithms.